

Structural Self-Energy Expansion (SSEE-V3.6)

A zero-free-parameter dark energy model from φ and π

Mike Edison Almeida Vallejo [ORCID: 0009-0008-2195-7836](https://orcid.org/0009-0008-2195-7836)
mike.almeida1721@gmail.com github.com/mikealmeida1721/SSEE

Seeking arXiv endorsement in astro-ph.CO (or gr-qc)

Core idea in one sentence

The core cosmological background — $w_0, w_a, H_0, \Omega_m, n_s$ — is derived *algebraically* from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π with **zero free parameters**. Perturbation variables (α_K, m_φ) are formulated as a working EFT proposal subject to full Boltzmann validation.

Key algebraic predictions

Quantity	SSEE algebraic value	Observed / constraint	Agreement
w_0	$-T_r/M_v = -0.8400$	DESI DR2: -0.827 ± 0.062	0.21σ
w_a	$-P_{sc}/K_v = -0.6699$	DESI DR2: $-0.75^{+0.29}_{-0.25}$	0.31σ
H_0	$3(\varphi + \pi)^2 = 67.96 \text{ km/s/Mpc}$	Planck 2018: 67.36 ± 0.54	1.11σ
r_d	CAMB+MIRA: 147.2 Mpc	Planck 2018: $147.09 \pm 0.26 \text{ Mpc}$	0.25σ
n_s	$1 - \varphi^{-7} = 0.9656$	Planck PR4: 0.9649 ± 0.0042	0.17σ
$\alpha_K(0)$	$3\Omega_{DE}(1 + w_\phi) = 0.073$	Euclid forecast sensitivity	< 0.1
m_φ	$\Sigma m_\nu \cdot H_0^{\text{alg}} = 5.71 \text{ eV}$	KATRIN/PTOLEMY (2027–30)	Prediction
k_{fs}	0.493 h/Mpc	DESI Y3/Euclid (2026–28)	Prediction

Seven-paper series (all reproducible)

1. **Framework + EFT**: Algebraic derivation of all constants; Bellini–Sawicki α -functions; α -attractor $\alpha = \varphi^4/3$; $\beta_c = -\text{AURA} = -3.998$ (plateau test). [22 pp]
2. **Bayesian MCMC**: 100 walkers $\times$ 25 000 steps vs. DESI DR2 + Planck + 7 clusters; $\Delta\text{BIC} = -5.55$ (SSEE favoured, $k_{\text{SSEE}} = 1$ vs. $k_{\Lambda\text{CDM}} = 3$, dynamic sector). [21 pp]
3. **CMB confrontation**: CAMB + MIRA two-sector mapping; full Cobaya+plik_lite MCMC; $H_0 = 67.08 \pm 0.03$; $\Delta\text{BIC} = -31.3$ (SSEE favoured under $k = 2$ model comparison). [18 pp]
4. **Nine Sovereignties**: All CMB observables ($\ell_1, \ell_2, \ell_3, \theta_*, A_s, n_s, \tau, r$) bounded algebraically. [15 pp]
5. **Israel–Stewart perturbations**: Causal bulk viscosity; $c_s^2 = 0$ exact; MIRA is background-sector effect (not perturbative); $\gamma_{\text{IS}} = 0.554 \approx \gamma_{\Lambda\text{CDM}}$. [24 pp]
6. **φ -DM two-sector**: $m_\varphi = 5.71 \text{ eV}$ algebraic; $\Omega_{\varphi\text{DM}} = (\text{MIRA} - 1)\Omega_{m,\text{dyn}}$; proposed resolution of $f\sigma_8$ tension to 0.50σ . [16 pp]
7. **Canonical EFT**: $\beta_c = -\text{AURA}$ (numerical -3.990 , gap 0.2% attributed to simplified background); $\alpha_T = \alpha_M = \alpha_B = 0$ exact (GW170817 consistent); $\alpha_K = 0.073$ algebraic; fundamental parameters λ, V_0, M proposed algebraically. [14 pp]

Falsifiability (explicit)

- $m_\varphi = 5.71 \text{ eV}$ — KATRIN (2027) or PTOLEMY (2030) will confirm or refute.
- $k_{fs} = 0.493 \text{ h/Mpc}$ — DESI Y3 + Euclid lensing will confirm or refute.
- $\alpha_K \approx 0.073$ — Euclid weak lensing will probe this kineticity limit.
- Any deviation of w_0 from -0.840 by $> 3\sigma$ refutes the framework.

What I am asking for

An arXiv endorsement in `astro-ph.CO` (or `gr-qc`) to submit Papers 1–3 as the first tranche. All code, data, and L^AT_EX sources are public at github.com/mikealmeida1721/SSEE under Apache-2.0.

I am happy to answer any technical questions or share the PDFs directly. Thank you for considering this request.

Code: Python (emcee, CAMB, CLASS), all reproducible in <30 min. **Data:** DESI DR2 (public), Planck PR4 (public), KiDS-1000 (public).